Card Classification Using Deep Learning

DUAL-MODEL APPROACH WITH

MOBILENETV2 & DENSENET-201

Team Introduction

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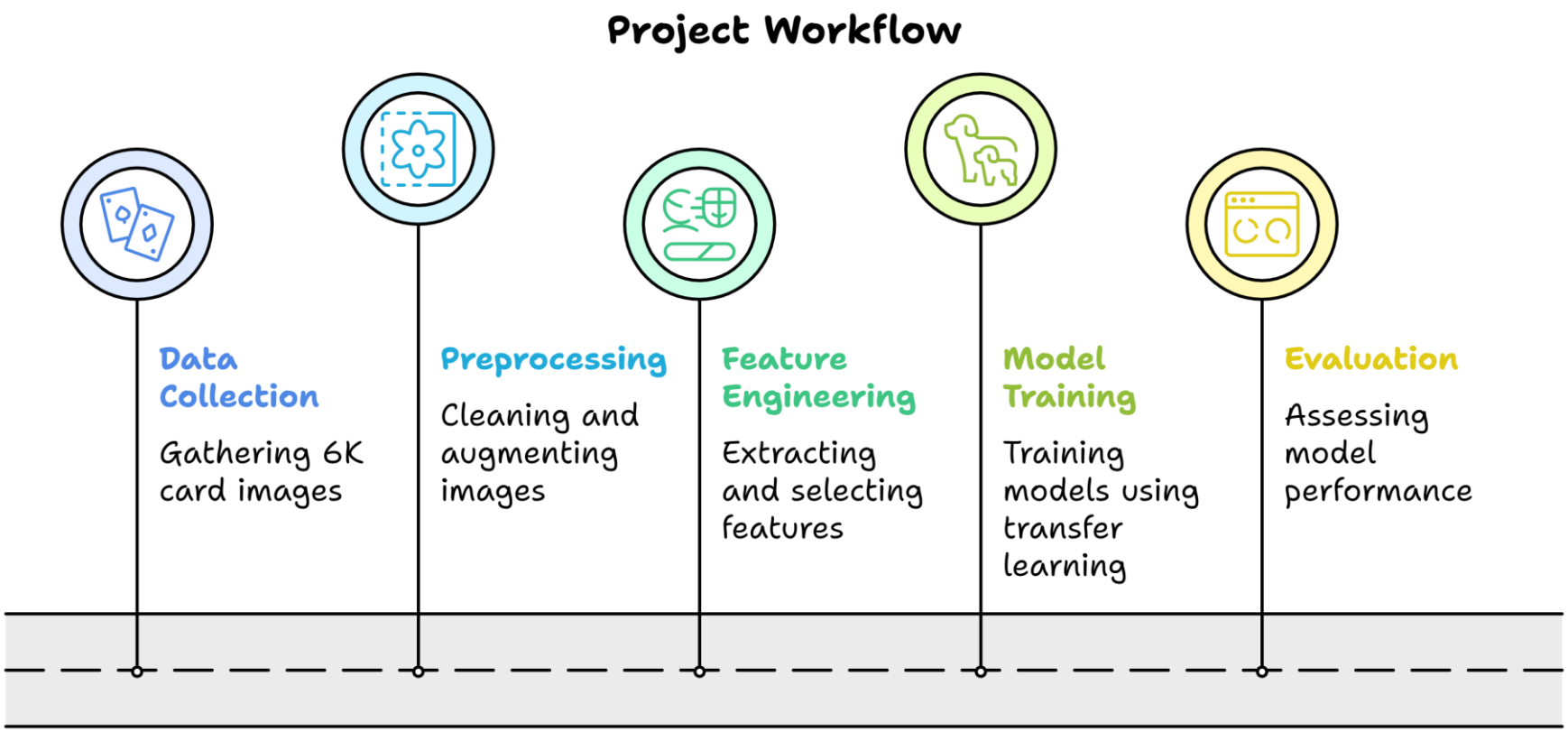
Introduction

- 🔍 **Project Importance:**
- • Automated card recognition for gaming industry
- • Accessibility tools for visually impaired players
- • Casino automation and security systems
- • AI-powered card game analysis

- 🎯 **Real-world Applications:**
- • Mobile gaming applications
- • Educational tools for card games
- • Fraud detection in gambling establishments
- • Digital card game assistants

Project Workflow



Dataset Description

- **Cards Image Dataset**

- **Dataset Statistics:**
- • Total Classes: 53 playing card categories
- • Image Size: 224×224×3
- • Data Split: Train (70%) / Validation (15%) / Test (15%)

- **Key Features:**
- • Balanced class distribution
- • Multiple orientations and lighting
- • Real-world card variations

CNN Architecture Overview

Feature	MobileNetV2 Stream	DenseNet-201 Stream
Base	MobileNetV2 Base	DenseNet201 Base
Global Pooling	GlobalAveragePooling2D	GlobalAveragePooling2D
Dense Layer	Dense(128)	Dense(256)
Output	Output(53)	Output(53)

Data Preprocessing Pipeline

🛠**Comprehensive Preprocessing**

- **Augmentation Techniques:**
- • Horizontal/Vertical flips
- • Rotation (±20°)
- • Zoom (80-120%)
- • Brightness adjustment

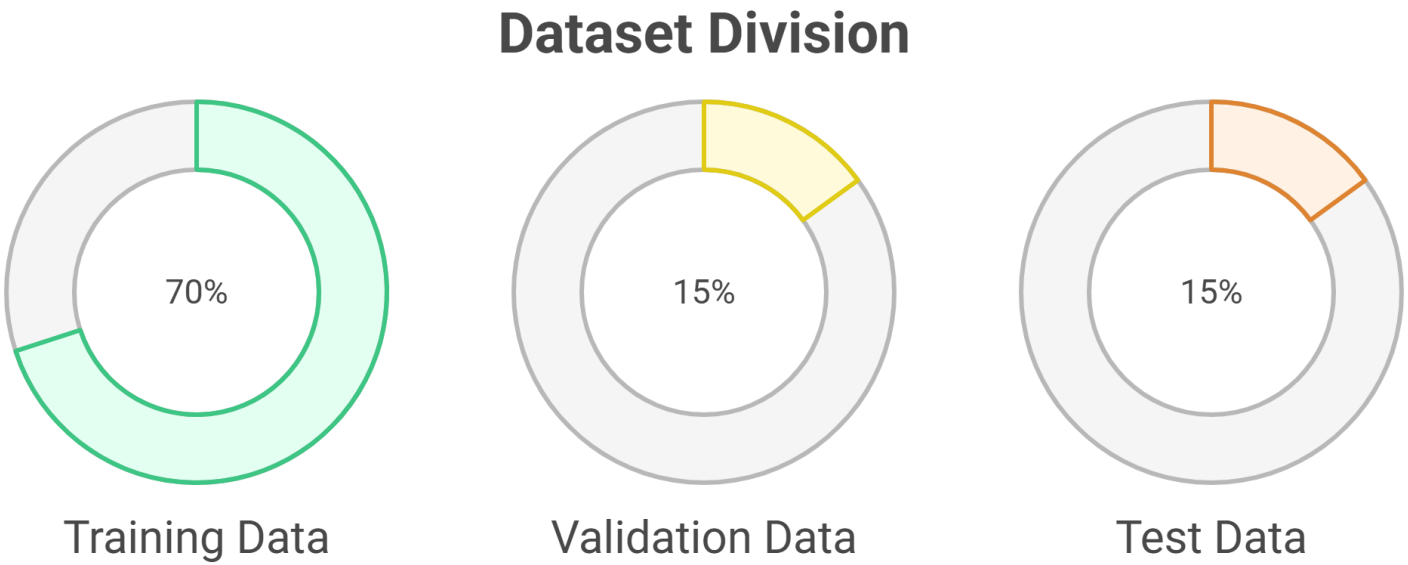
- **Filter Applications:**
- • Median & Gaussian blur
- • Edge detection (Canny)
- • Sharpening filters
- • Embossing effects

Hyperparameter Optimization

- **Parameters:**
- • Learning Rate: 1e-4
- • Batch Size: 32
- • Epochs: 20-25
- • Dropout Rate: 0.5

- **Training Configuration:**
- • Optimizer: Adam
- • Loss: Categorical Crossentropy
- • Callbacks: EarlyStopping, ReduceLROnPlateau

Data Segmentation Strategy



Model Training & Evaluation

🎯 **Training & Evaluation Framework**

- **Evaluation Metrics:**
- • Training/Validation Accuracy
- • Training/Validation Loss
- • Test Accuracy
- • Confusion Matrix
- • ROC Curves

- **Model Comparison:**
- • MobileNetV2 vs DenseNet-201
- • Feature extraction capabilities
- • Convergence behavior

Results Analysis

Accuracy Comparison of Models

Model Name	Train Accuracy	Validation Accuracy	Test Accuracy
MobileNetV2	92.13%	86.04%	85.28%
DenseNet-201	85.66%	80.00%	73.96%

Classification Results & Analysis

- **Model Predictions:**
- • Side-by-side model comparisons
- • External image testing

- **Success Metrics:**
- • High precision across all card classes
- • Consistent performance on test set
- • Good generalization to unseen data

External Image Testing

MobileNetV2: king of spades

MobileNetV2: queen of clubs

MobileNetV2: ace of spades

MobileNetV2: queen of spades

MobileNetV2: king of clubs

DenseNet-201: king of spades

DenseNet-201: queen of clubs

DenseNet-201: ace of spades

DenseNet-201: queen of spades

DenseNet-201: king of clubs

Conclusion & Future Work

- **Project Achievements :**
- • Successful dual CNN architecture
- • Comprehensive preprocessing pipeline
- • Effective transfer learning
- • High-accuracy card classification

- **Future Directions:**
- • Real-time video processing
- • Mobile application development
- • Multi-card hand recognition

Thank You!

WE APPRECIATE YOUR ATTENTION AND TIME.